

# Astrodynamics I: Final Exam

June 4, 2015 10:00-11:45PM (105 minutes)

**Closed Books/Notes, No Internet/Communication Devices (including Cell Phones)**

1. (15) A spacecraft is orbiting a planet in a circular orbit with a radius of 4 Planet Radii ( $a=4, e=0$ ). A maneuver is performed that rotates the velocity vector in the orbital plane by an angle of  $45^\circ$  towards the planet.
  - a. (5) Find the new semimajor axis  $a_n$  and the new eccentricity  $e_n$  following this maneuver.
  - b. (5) At what point in its new orbit is the spacecraft in terms of true anomaly ( $\nu$ )?
  - c. (5) Will the spacecraft impact the planet? Justify your answer.

2. (20) Conversion of Orbital Elements to state vectors

- a. (6) Show that the position and velocity vector in the perifocal coordinate system is:

$$\vec{r}_{PQW} = \begin{bmatrix} \frac{pcos\nu}{1+ecos\nu} \\ \frac{psin\nu}{1+ecos\nu} \\ 0 \end{bmatrix} \text{ and } \vec{v}_{PQW} = \begin{bmatrix} -\sqrt{\frac{\mu}{p}} \sin\nu \\ \sqrt{\frac{\mu}{p}} (e + \cos\nu) \\ 0 \end{bmatrix}$$

- b. (6) For the following orbital elements:

$$p = 0.23DU, e = 0.82, i = 90^\circ, \Omega = 180^\circ, \omega = 260^\circ, \nu = 190^\circ$$

determine  $\vec{r}_{PQW}$  and  $\vec{v}_{PQW}$ . Note: set  $\mu=1$ .

- c. (8) By a suitable coordinate transformation, express  $\vec{r}$  and  $\vec{v}$  in the geocentric equatorial system in the  $\hat{I}, \hat{J}, \hat{K}$  axis.

3. (10) Prove that the eccentricity vector  $\vec{e} = \frac{1}{\mu} \left[ \left( v^2 - \frac{\mu}{r} \right) \vec{r} - (\vec{r} \cdot \vec{v}) \vec{v} \right]$  always points to the periapsis.

**To be continued on the next page.....**

4. (20) State the definition of following terminologies SUCCINCTLY and PRECISELY:
  - a. (4) Kepler's Problem
  - b. (4) Geodetic Latitude & Astronomical Latitude
  - c. (4) Satellite Coordinate System (denoted as RSW) = Gaussian Coordinate System
  - d. (4) Universal Time
  - e. (4) Equation of Equinoxes
5. (15) Use the flow chart discussed in class to briefly describe the coordinate transformation process between Celestial (=FK5=J2000) and Terrestrial (=ITRF) coordinates.
6. (20) Given  $\vec{r}_{IJK}, \vec{v}_{IJK}, \vec{r}_{siteIJK}, \Phi_{gd}, \theta_{LST}$ , describe the procedure to compute range ( $\rho$ ), azimuth ( $\beta$ ), elevation ( $el$ ), and their time rates.